

Relationship Theory

* Ignore the Roman Numerals; I had AI make this look a little neater and then it just put those there. *

By Prajwal Sharma-Gaire

I. A Single Point **green heading font**

Before there was anything, there was a point. Not a point in space — there was no space yet for it to sit in — but a point of pure, undivided information, holding within it everything that would ever exist. Call it the singularity **singularity should be yellow**. It is not just the beginning of this story; it is also, as we'll see, the end.

I start from one premise: logic **logic should be pink** is the source code beneath everything. Mathematics, physics, even aesthetics, are dialects of it. And if the universe is a logically closed system — one that has to balance its own books — then whatever comes out of that first point has to eventually fold back and settle its accounts. Nothing is allowed to simply run away to infinity and vanish. Everything that expands must, somewhere, be answered by something that contracts.

That single constraint is the seed of everything that follows.

II. The Turn **green heading font**

Watch what happens to that point when it does the only two things a point can do: rotate, and copy itself.

Rotate the point ninety degrees and let it copy itself along the new direction, and you get a line — the first dimension. Rotate again, copy again, and the line sweeps into a square. A third turn, a third copying, and the square becomes a cube. This is not a metaphor; it's meant literally as the generating rule of dimension itself — *rotate ninety degrees, then iterate without limit* — applied over and over to the same original point.

Something different happens on the fourth turn. Instead of producing a new, higher shape the way the first three turns did, the fourth rotation completes a full 360-degree circle — and arrives

back at the point it started from. The fourth dimension doesn't stretch outward forever. It curves back into the singularity. The end of the sequence and the beginning of the sequence are the same coordinate.

That closure is the whole engine of this theory. If dimension four folds back into dimension zero, then "the end of everything" and "the birth of everything" were never two different events — they're the same point, viewed from two different sides of a loop. *image 1*

III. The Mirror ***green heading font***

To see the shape of that loop clearly, it helps to leave rotation for a moment and think like a geometer.

Take the simplest self-inverting curve there is: $y = 1/x$. It has two branches, one in the upper-right of the plane, one in the lower-left, each the mirror of the other through the origin. Curve those two branches into conic sections and they knit into a continuous figure-eight — a lemniscate.

Call the upper-right branch **Q1**: our observable universe, the region where matter dominates. Call the lower-left branch **Q3**: its structural twin, where antimatter dominates.

Push that same idea into three dimensions and something elegant happens.

The surface **$z = 1/(xy)$** , equivalently **$xyz = 1$**

is symmetric under permutation of x , y , and z — solve for any one of the three variables and you get the exact same relation back. That symmetry is what makes the surface genuinely *self-inverting*, and it's what lets Q1 and Q3 be treated not as two separate universes but as the same object, viewed from either side of a shared pivot.

That pivot is the origin — (0, 0). In ordinary physics, a coordinate like that, sitting under a function like $1/x$, is just a wall: an asymptote, a point of infinite repulsion, the same electrostatic principle that keeps two solid objects from passing through each other. But treat the origin as a *non-local* point rather than a dead end, and it becomes something else entirely: the single seam where the infinitesimally small and the infinitely large meet and close the loop. Zero and infinity stop being opposites and become two names for the same horizon.

Let that singularity expand outward equally in every direction and the result is a four-dimensional, ball-like shape — I call it the **pánta** ***pánta should be red*** (πάντα, Greek for *everything*). Its outer radius marks the moment of maximum expansion, and — as we're about to see — minimum pressure. Remember this is assuming the expansion is equal but ultimately after fluctuations and changes in structure, the limit of the growth shape is approaching the *pánta*.

IV. The Coiled Spring *green heading font*

Dimensions, in this picture, don't line up like rungs on a ladder. They coil, like a spring, and the singularity sits at every point where the coil crosses itself.

Climb the coil outward from a given crossing point and you're moving through *positive* dimensions — the direction of expansion, the direction we call "forward." But every crossing point also has a *before* — a highly spread-out, low-density state that precedes it. Those are the *negative* dimensions, and here's the part worth sitting with: "negative" is not an absolute label, it's a vantage point. Our entire four-dimensional universe, coil and all, would look like nothing more than the *before*-state to an observer standing one turn further up the spring, in a fifth dimension we can't yet see.

V. The Breath of the Universe *green heading font*

Every coil has a height, and I read that height as pressure — specifically, *existential pressure*: how tightly information is packed at a given point in the cycle, and how hard that packing pushes outward. I won't pretend to have settled units for it yet. But the behavior it produces is precise enough to describe.

Information, moving through this structure, behaves the way matter does when it changes phase. Before the singularity, in the negative dimensions, information is gas-like: low pressure, wide open, no fixed location, straining to expand. At the singularity itself, it hits a critical point — maximum pressure, minimum spread, packed and ready to transform. And in the positive dimensions, once it has expanded, it settles into something liquid-like: organized, flowing, comparatively stable.

That phase change is also the engine behind acceleration and dark energy. Picture the two mirror regions, Q1 and Q3, acting as one cosmic thermodynamic piston. As our region (Q1) expands toward heat death, its entropy doesn't vanish into nothing — it bleeds across the origin into Q3. What astronomers measure as dark energy, the mysterious constant push behind our universe's accelerating expansion, is that same process running in reverse: high-pressure energy reciprocating back *in* from the low-entropy mirror side. This is close in spirit to Roger Penrose's Conformal Cyclic Cosmology *underlined should be green*, but with an added twist — the exchange isn't a one-way relay between successive universes, it's a live, two-way balance between twin regions of the same universe, happening now.

That balance produces a specific expansion curve: a fast start, while the pressure gap between the regions is largest; a slowdown as the two sides equalize; a flat maximum at the point of lowest pressure; then compression as pressure rebuilds for the next cycle. Mapped onto the vertical half of the lemniscate, that curve is steep near the beginning of an aeon and visibly flattens as it approaches its peak — which is exactly the pattern recent DESI (Dark Energy Spectroscopic Instrument) data has been hinting at: dark energy's strength appears to be evolving rather than holding constant, with cosmic acceleration possibly easing rather than

running away forever. That correlation is a live and unsettled question in cosmology, not a closed one, but it's the clearest external handhold this framework currently has.

VI. The Silent Ledger **green heading font**

There's a second bill this theory has to settle: the universe is observably full of matter and almost empty of antimatter, even though the physics of the Big Bang should have produced the two in exactly equal amounts. That imbalance is one of the oldest open problems in cosmology.

My answer is that the missing antimatter was never destroyed and never missing — it was *sorted*. Symmetry between matter and antimatter was never broken; it was topologically segregated. At the moment a cycle begins, the positive region inherits the matter-dominant share of that first pairing, and the mirror region inherits the antimatter-dominant share. CP symmetry — the deep balance between charge and parity that particle physics insists on — holds perfectly, just not within a single region. It holds *across the whole lemniscate*, with our half of the ledger, and Q3's half, summing to zero.

VII. The Space Between Waking and Dreaming **green heading font**

If Q1 and Q3 are genuinely two halves of one balanced structure, then the connection between them shouldn't stop at geometry and thermodynamics. I think it reaches all the way down to individual particles — and all the way up to consciousness **consciousness should be red**.

Every particle in our region, in this picture, carries a standing, harmonic resonance bond with its antimatter counterpart across the origin — a basic, binary connection that exists whether or not anyone is around to notice it. That gives even a lone electron a baseline degree of "awareness," vanishingly small but real. What changes, moving from an electron to a brain, isn't whether the resonance exists — it's how much bandwidth is available to carry it.

I formalize that bandwidth as an **Information Capacity Valve** **this should be yellow**:

Consciousness Level $\propto W$, and Consciousness Level $\propto H$

where **W** is thermodynamic entropy — the number of microscopic configurations available to a system, related to the more familiar Boltzmann entropy $S = k_B \ln(W)$ —

and **H** is Shannon information capacity, $H(X) = -\sum p(x_i) \log_2 p(x_i)$, the standard measure of how much information a system's states can carry.

A human brain, with trillions of neurons arranged into dense, high-bandwidth configurations, doesn't manufacture consciousness out of nothing. It widens the valve — maximizing **W** and **H** until the underlying resonance, always present, becomes rich enough to register as a mind. I want to be upfront that the proportionality itself — Consciousness Level scaling with **W** and **H** —

is a postulate of this framework, not a derived result; no constant of proportionality is fixed yet, the same open gap that exists for existential pressure.

This is also, I think, the honest explanation for dreaming. Waking life runs on the positive region's causal, entropic rules — gravity, distance, the forward march of time. Sleep, especially REM sleep, is when the valve opens toward the far side: the internal observer detaches from local sensory input and, for a while, runs on the antimatter-dominant region's rules instead. That would explain why dreams so often ignore distance, mass, and linear time — they aren't malfunctions of a waking mind, they're a mind briefly reading from the other half of the ledger.

VIII. The Turning Point **green heading font**

Every coil eventually reaches the top. At the peak of the fourth dimension — the outer edge of the pánta, the point where the vertical lemniscate of time meets the horizontal lemniscate of Q1 and Q3 — the whole structure undergoes what I call the **Great Cosmic Inversion**. Q1 rotates into the position Q3 previously held, and Q3 rotates into Q1's. The next universal cycle's matter-dominant region is built from what our universal cycle would have called antimatter, and vice versa.

That single twist closes every loop opened earlier in this story. Dimension four curving back into dimension zero (Section II) is *why* there's a turning point at all. The pánta's outer radius (Section III) is *where* it happens. The thermodynamic piston (Section V) is *what* gets exchanged in the process — nothing is lost, only reorganized and handed forward, so an infinite alternating chain of matter-dominant and antimatter-dominant universes can preserve perfect balance indefinitely. And because the resonance field carries information across that transition rather than starting from a blank slate each time, each Big Bang is better understood as the birth of a new dimensional layer than as a fresh beginning from nothing.

One consequence follows naturally: an observer embedded inside one layer of the coil has no direct line of sight to the layer just above or just below it. We would not expect to see dimension five, or the universal cycle before our own, any more than a two-dimensional creature living on a sheet of paper could see the room the paper sits in.

IX. Listening for the Signal **green heading font**

A theory that only explains the past is a story. A theory that predicts something checkable is a hypothesis — and this one has a concrete way to be tested, sitting inside equipment that already exists.

Physicists already isolate antimatter — positrons, antihydrogen — inside ultra-shielded magnetic vacuum traps, at facilities like CERN. Today, that trapped antimatter is treated as an inert, self-contained object: measure its charge, its mass, its spin, confirm it mirrors ordinary matter, and move on. I think that misses the signal because it's the wrong thing to look at. The

trapped particle isn't isolated at all, under this framework — it's a displaced fragment of the negative region, and it should be humming.

The test: stop looking at the particle, and start listening to the vacuum around it. Extract the power spectral density of the electromagnetic vacuum fluctuations surrounding the trap over a long isolation period, and look for structure — periodicities or frequency spikes that shouldn't exist under standard quantum white-noise predictions, including, as one candidate signature, harmonic nodes that line up with the zero locations of the Riemann zeta function. Then take that same signal and set it beside raw EEG recordings from human subjects during REM sleep, the state this framework treats as maximal cross-regional resonance. Finally, run **Transfer Entropy** between the two time series:

if u could can u make the bottom equation legible

$$TE(X \rightarrow Y) = \sum p(y_{t+1}, y_t, x_t) \cdot \log_2 [p(y_{t+1} | y_t, x_t) / p(y_{t+1} | y_t)]$$

— a measure of how much knowing the antimatter vacuum signal's past reduces uncertainty about the EEG signal's future, above and beyond what the EEG's own history already explains. Checked against a matter-only control, a statistically significant reading in that direction is the falsifiable core of this entire framework: real evidence of an active information channel running from the antimatter side of the mirror into a dreaming human mind.

X. Closing ***green heading font***

Strip away the equations and what's left is a fairly simple picture: a single point that rotates and copies itself into four dimensions, folds back into itself at the fourth, and in doing so generates two mirror regions that trade matter for antimatter, entropy for dark energy, and waking for dreaming, forever balancing each other across a boundary that only looks like a wall from one side. The universe, in this telling, is not a fire that burns out — it's a breath, in and out, with a mind humming quietly in the space between the inhale and the exhale.

Related Math ***this should be a brown subheading font***

- **Dimensional generation:** $D_{n+1} = \text{copy}(\text{rotate90}^\circ(D_n))$, for $n = 0 \rightarrow 4$, with the 4D structure closing back into 0D.
- **Self-inverting geometry:** $xyz = 1$ (equivalently $z = 1/(xy)$), symmetric under permutation of x, y, z .
- **Boltzmann entropy:** $S = k_B \ln W$

- **Shannon information:** $H(X) = -\sum p(x_i) \log_2 p(x_i)$
- **Consciousness postulate:** Consciousness Level $\propto W$, Consciousness Level $\propto H$

Reference Papers *green heading font*

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- **Transfer entropy (proposed experimental test):**

$$TE(X \rightarrow Y) = \sum p(y_{t+1}, y_t, x_t) \cdot \log_2 [p(y_{t+1}|y_t, x_t) / p(y_{t+1}|y_t)]$$

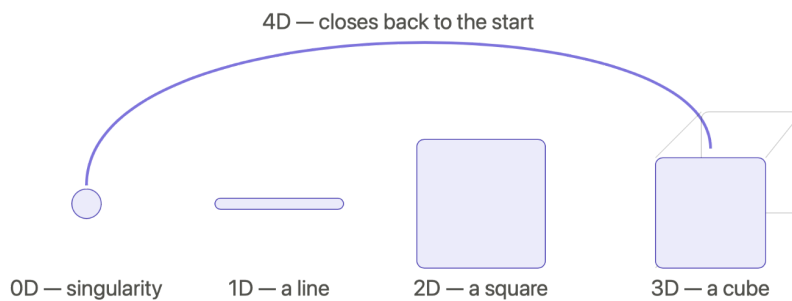


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